



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

CSA0618 – DESIGN AND ANALYSIS OF ALGORITHM

EXPERIMENT NO: 03

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Program:

```
def factorial_iterative(n):  
    fact = 1  
    for i in range(1, n + 1):  
        fact *= i  
    return fact  
  
def factorial_recursive(n):  
    if n == 0 or n == 1:  
        return 1  
    return n * factorial_recursive(n - 1)  
  
n = 5  
  
print("Iterative Factorial:", factorial_iterative(n))  
print("Recursive Factorial:", factorial_recursive(n))
```

Output:

main.py	Visualize	Run ▶	Output
<pre>1 def factorial_iterative(n): 2 fact = 1 3 for i in range(1, n + 1): 4 fact *= i 5 return fact 6 7 8 def factorial_recursive(n): 9 if n == 0 or n == 1: 10 return 1 11 return n * factorial_recursive(n - 1) 12 n = 5 13 14 print("Iterative Factorial:", factorial_iterative(n)) 15 print("Recursive Factorial:", factorial_recursive(n))</pre>			<pre>Iterative Factorial: 120 Recursive Factorial: 120</pre>